

High Performance Programming

Lecture:
“GPU set-up”



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We will work with GPUs in different ways

CUDA:

Good: Low complexity (beginners)

Downsides: NVIDIA exclusive and low level of detail in the programming.

Check with `nvidia-smi --list-gpus` and search in NVIDIA webpage [here](#)

PyOpenCL: Hardcore GPU programming

Good: Full control on partitioning, mapping, etc.

Downsides: Difficult to get to work.

GPU programming with Nvidia CUDA toolkit

If your GPU supports CUDA

1. Install drivers (if missing)

2. Install anaconda

3. In a conda environment with `conda create --name env_name`

a. Install **cuda toolkit** in a conda environment

b. Check installation running: **numba -s**

c. The driver and runtime version must match



Outside of conda environment (no cudatoolkit)

__CUDA Information__

CUDA Device Initialized : True

CUDA Driver Version : (11, 6)

CUDA Runtime Version : ?

Inside conda environment

__CUDA Information__

CUDA Device Initialized : True

CUDA Driver Version : (11, 6)

CUDA Runtime Version : 11060

GPU programming with Nvidia CUDA (alt. set-up)

If your GPU supports CUDA

1.Install drivers (if missing)

2.Install PyTorch

3.Check installation in Python:

```
>>> torch.cuda.is_available()
```

```
True
```

```
>>> torch.cuda.current_device()
```

```
0
```

```
>>> torch.cuda.get_device_name()
```

```
'NVIDIA GeForce 940MX'
```



PyOpenCL and GPU drivers

Test installation by running

```
import pyopencl as cl  
print(cl.get_platforms())
```

My output

```
[<pyopencl.Platform 'Oclgrind' at 0x21e8860>,  
<pyopencl.Platform 'NVIDIA CUDA' at 0x223bb40>,  
<pyopencl.Platform 'Portable Computing Language' at 0x7f9025e29320>]
```

Oclgrind:

Platform to enable the creation of developer tools for analysis and debugging of OpenCL programs. Simulates how OpenCL kernels execute. More info [here](#)

Choose platform

Installing PyOpencl

Installation guide <https://document.tician.de/pyopencl/misc.html>

And <https://pypi.org/project/pyopencl/>

The process is “straightforward” on Linux

“...analogous steps for Windows **should** also work”

Try with windows binaries <https://www.lfd.uci.edu/~gohlke/pythonlibs/#pyopencl>



OpenCL